

✓ Task: Data Visualization Project

Objective:

Transform raw sales data into meaningful visualizations using Python libraries like Matplotlib and Seaborn.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab

✓ Dataset Description

The dataset contains monthly business performance data:

Columns:

- Month: Sales month
- Sales: Total sales amount
- Customers: Number of customers
- Profit: Monthly profit

The goal is to analyze trends and identify business insights.

import libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Style
sns.set_style("whitegrid")

print("Libraries imported successfully")
```

```
Libraries imported successfully
```

load data

```
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [25000, 30000, 28000, 45000, 50000, 60000],
    "Customers": [120, 150, 140, 200, 240, 300],
    "Profit": [5000, 7000, 6500, 12000, 15000, 20000]
}

df = pd.DataFrame(data)

df
```

	Month	Sales	Customers	Profit
0	Jan	25000	120	5000
1	Feb	30000	150	7000
2	Mar	28000	140	6500
3	Apr	45000	200	12000
4	May	50000	240	15000
5	Jun	60000	300	20000



Next steps: [Generate code with df](#) [New interactive sheet](#)

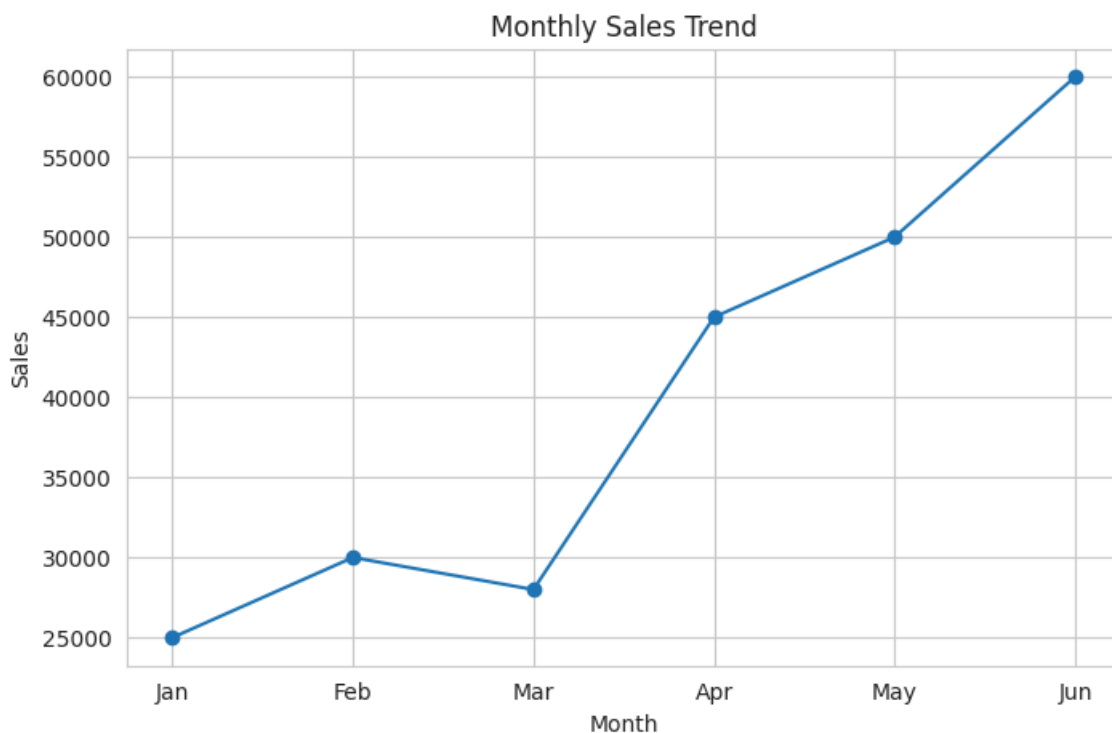
data visualization

```
plt.figure(figsize=(8,5))

plt.plot(df["Month"], df["Sales"], marker="o")

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```

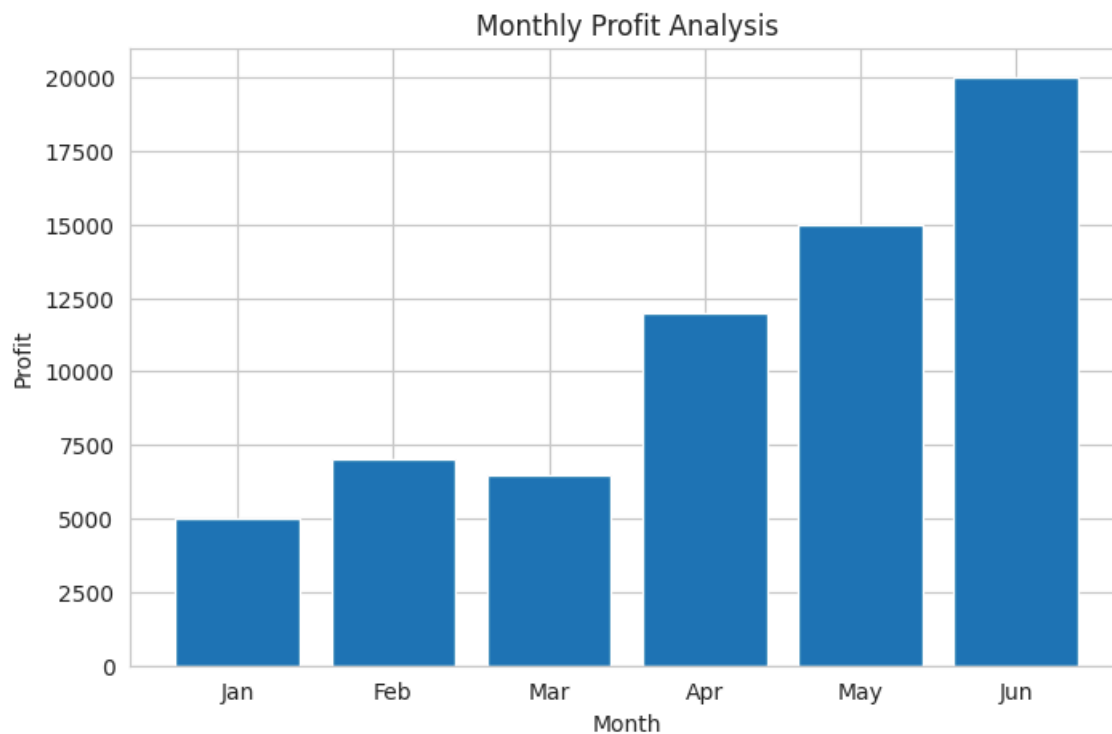


```
plt.figure(figsize=(8,5))

plt.bar(df["Month"], df["Profit"])

plt.title("Monthly Profit Analysis")
plt.xlabel("Month")
plt.ylabel("Profit")

plt.show()
```

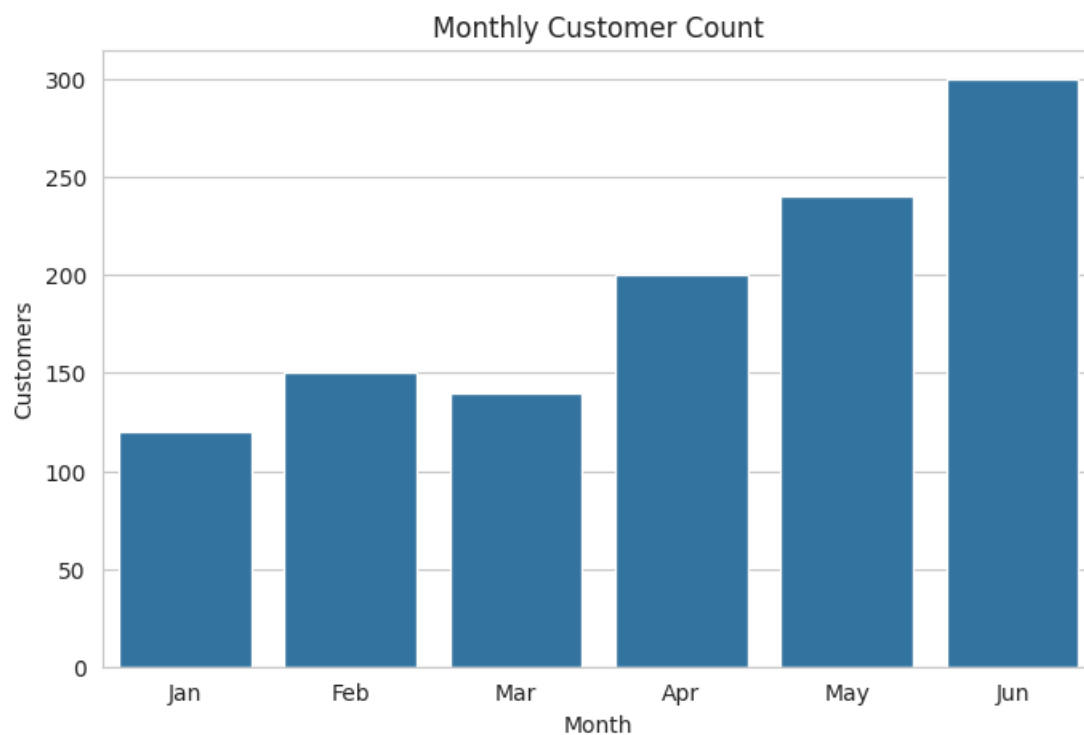


```
plt.figure(figsize=(8,5))

sns.barplot(
    x=df["Month"],
    y=df["Customers"]
)

plt.title("Monthly Customer Count")

plt.show()
```



```
plt.figure(figsize=(8,5))

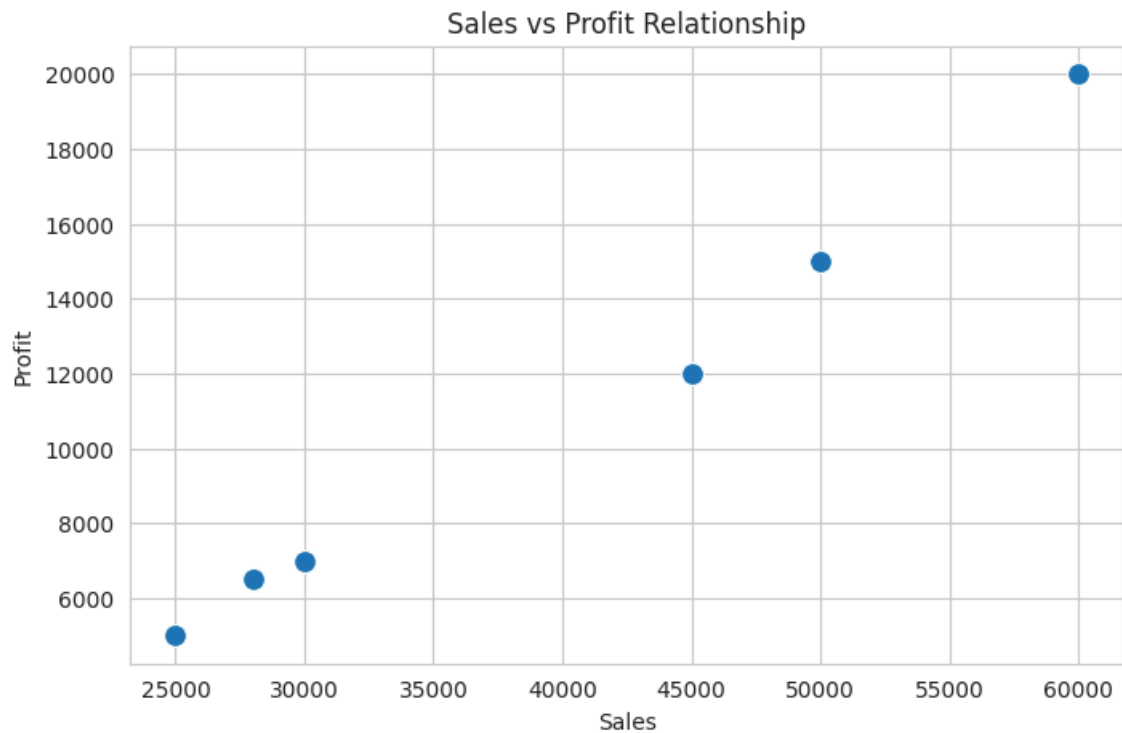
sns.scatterplot(
    x=df["Sales"],
```

```
y=df["Profit"],
s=100
)

plt.title("Sales vs Profit Relationship")

plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()
```

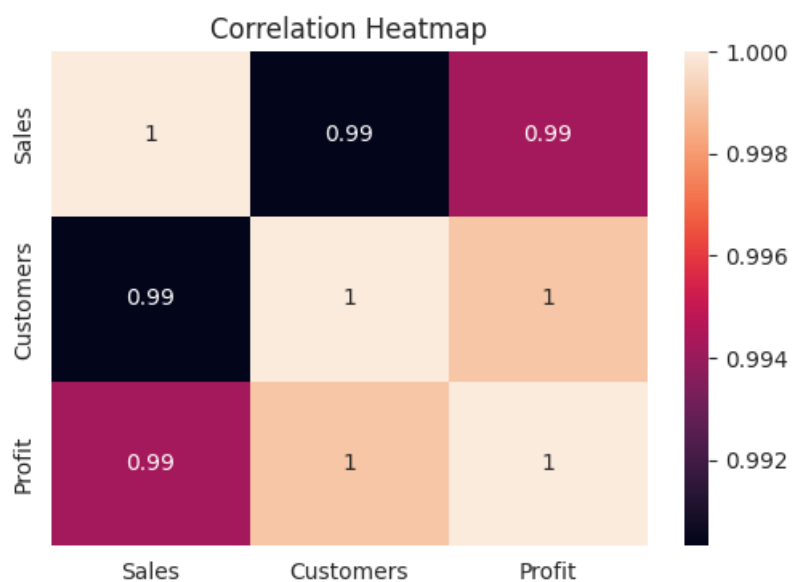


```
plt.figure(figsize=(6,4))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```



```
fig, axes = plt.subplots(2,2, figsize=(12,8))

# Sales
axes[0,0].plot(df["Month"], df["Sales"], marker="o")
axes[0,0].set_title("Sales Trend")

# Profit
axes[0,1].bar(df["Month"], df["Profit"])
axes[0,1].set_title("Profit Analysis")

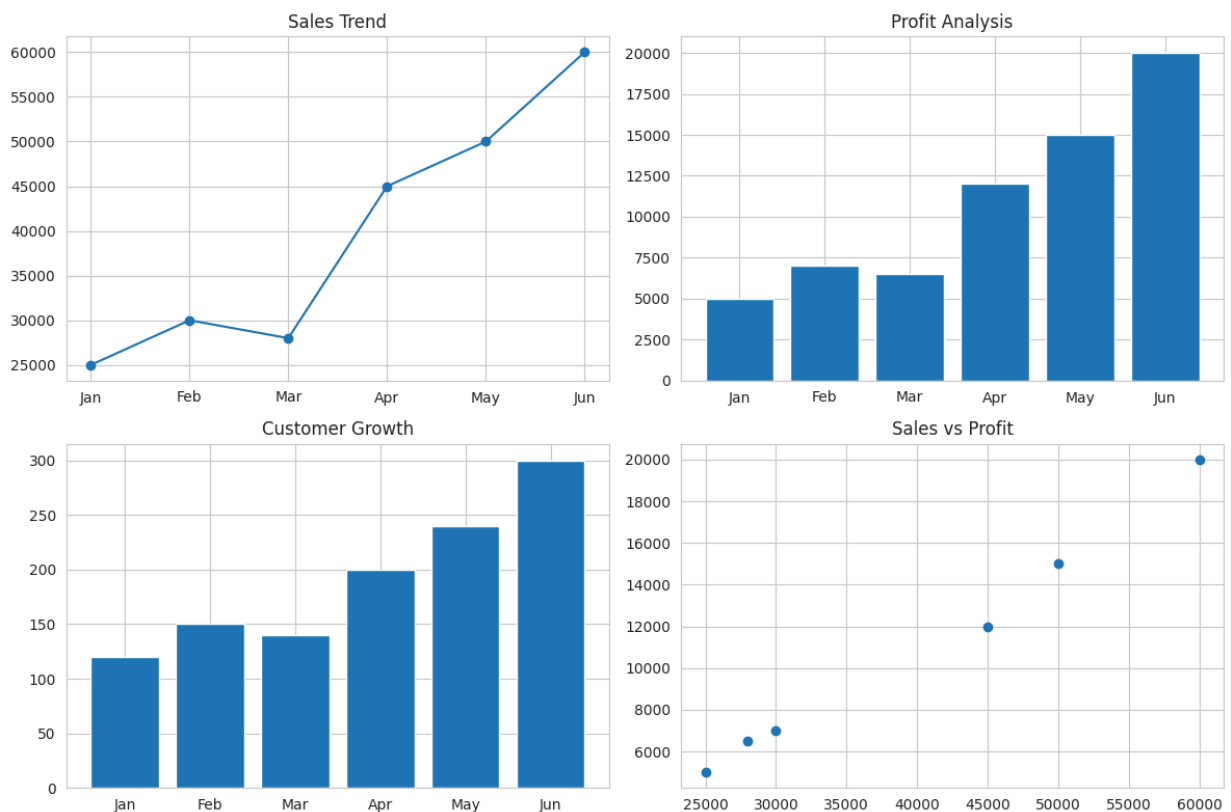
# Customers
axes[1,0].bar(df["Month"], df["Customers"])
axes[1,0].set_title("Customer Growth")

# Sales vs Profit
axes[1,1].scatter(df["Sales"], df["Profit"])
axes[1,1].set_title("Sales vs Profit")

plt.tight_layout()

# Save the figures before showing them
plt.savefig("Business_Dashboard.png")
plt.savefig("sales_dashboard.png")

plt.show()
```



insights

✓ Data Insights

1. Sales increased continuously from January to June.
2. Profit growth followed the sales trend.
3. Customer count increased every month.
4. Sales and profit have a positive relationship.

5. The best performing month was June.

dashboard

```
plt.savefig("sales_dashboard.png")
```

<Figure size 640x480 with 0 Axes>

```
plt.tight_layout()
```

<Figure size 640x480 with 0 Axes>

Conclusion

This analysis shows that sales, customers, and profit are positively related. The business shows steady growth from January to June, with June being the best-performing month.